CANTEEN FOOD WASTE PREDICTION & TRACKER

Using Arrays, Sorting and Searching

IT23302 - Data Structures Project

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C PROGRAM SOURCE CODE

```
#include <stdio.h>
#include <string.h>
#include <stdlib.h>

#define MAX 100

struct FoodRecord {
    char itemName[50];
    char date[15];
    int prepared;
    int consumed;
    int wasted;
    float wastePercentage;
};

struct FoodRecord records[MAX];
int count = 0;

void calculateWaste(int i) {
    records[i].wasted = records[i].prepared - records[i].consumed;
    if (records[i].prepared > 0) {
        records[i].wastePercentage = ((float)records[i].wasted / records[i].prepared) * 100;
    } else {
        records[i].wastePercentage = 0;
    }
}

void addRecord() {
    if (count >= MAX) {
        printf("Storage is full!\\n");
        return;
    }
    printf("Enter Item Name: ");
    scanf("%s", records[count].itemName);
    printf("Enter Date (DD-MM-YYYY): ");
    scanf("%s", records[count].date);
    printf("Enter Quantity Prepared: ");
    scanf("%d", &records[count].prepared);
    printf("Enter Quantity Consumed: ");
    scanf("%d", &records[count].consumed);

    calculateWaste(count);
    count++;
    printf("Record added successfully!\\n");
}

void showReport() {
    if (count == 0) {
        printf("No records found.\\n");
        return;
    }
    printf("\\n--- Canteen Food Waste Report ---\\n");
    printf("%-15s %-12s %-10s %-10s %-10s %-10s\\n", "Item", "Date", "Prepared", "Consumed",
"Wasted", "Waste %");
    for (int i = 0; i < count; i++) {
        printf("%-15s %-12s %-10d %-10d %-10d %-9.2f%%\\n",
            records[i].itemName, records[i].date, records[i].prepared,
            records[i].consumed, records[i].wasted, records[i].wastePercentage);
    }
}

void predictQuantity(char* name) {
    int totalPrepared = 0, totalConsumed = 0, matches = 0;
    for (int i = 0; i < count; i++) {
        if (strcmp(records[i].itemName, name) == 0) {
            totalPrepared += records[i].prepared;
            totalConsumed += records[i].consumed;
            matches++;
        }
    }
    if (matches > 0) {
        int avgConsumed = totalConsumed / matches;
        printf("\\nPrediction for %s: Suggested quantity to prepare tomorrow is approximately %d
units based on past consumption.\\n", name, avgConsumed + 5);
    } else {
        printf("Item not found for prediction.\\n");
    }
}
```

```
void searchRecord() {
    char target[50];
    printf("Enter Item Name to search: ");
    scanf("%s", target);
    int found = 0;
    for (int i = 0; i < count; i++) {
        if (strcmp(records[i].itemName, target) == 0) {
            printf("Found: Item: %s | Date: %s | Prepared: %d | Wasted: %d\\n",
                records[i].itemName, records[i].date, records[i].prepared, records[i].wasted);
            found = 1;
        }
    }
    if (!found) {
        printf("Record not found!\\n");
    }
}

void sortRecords() {
    struct FoodRecord temp;
    for (int i = 0; i < count - 1; i++) {
        for (int j = 0; j < count - i - 1; j++) {
            if (records[j].wasted < records[j + 1].wasted) { // Descending order of waste
                temp = records[j];
                records[j] = records[j + 1];
                records[j + 1] = temp;
            }
        }
    }
    printf("Records sorted by highest waste successfully!\\n");
}

void saveToFile() {
    FILE *file = fopen("canteen_data.txt", "w");
    if (!file) {
        printf("Error opening file for saving!\\n");
        return;
    }
    for (int i = 0; i < count; i++) {
        fprintf(file, "%s %s %d %d %d %f\\n",
            records[i].itemName, records[i].date, records[i].prepared,
            records[i].consumed, records[i].wasted, records[i].wastePercentage);
    }
    fclose(file);
    printf("Data saved to file successfully!\\n");
}

void loadFromFile() {
    FILE *file = fopen("canteen_data.txt", "r");
    if (!file) return;
    count = 0;
    while (fscanf(file, "%s %s %d %d %d %f\\n",
        records[count].itemName, records[count].date, &records[count].prepared,
        &records[count].consumed, &records[count].wasted, &records[count].wastePercentage) == 6) {
        count++;
    }
    fclose(file);
}

int main() {
    loadFromFile();
    int choice;
    char queryName[50];

    while (1) {
        printf("\\n--- CANTEEN FOOD WASTE TRACKER ---\\n");
        printf("1. Add Record\\n");
        printf("2. Show Report\\n");
        printf("3. Sort by Waste (Highest First)\\n");
        printf("4. Search Record by Item Name\\n");
        printf("5. Predict Next Day Quantity\\n");
        printf("6. Save & Exit\\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1: addRecord(); break;
            case 2: showReport(); break;
            case 3: sortRecords(); break;
            case 4: searchRecord(); break;
            case 5:
                printf("Enter item name for prediction: ");
                scanf("%s", queryName);
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                predictQuantity(queryName);
                break;
            case 6:
                saveToFile();
                exit(0);
            default: printf("Invalid choice! Try again.\\n");
        }
    }
    return 0;
}
```